

Software Project Tracking

Hands-on with Trello and JIRA

CA Firm Attendance and Article Management System

Assignment No. 10

Group: E-17

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1. Problem Statement

Hands-on with Project Management Tools: Practice project tracking, task assignment, and progress monitoring using Trello and JIRA on a simulated project.

2. Introduction

Modern software development requires robust tools to manage complexities, coordinate team members, and ensure timely delivery. Agile project management relies heavily on tracking systems to maintain visibility over tasks, bugs, and sprints.

In this assignment, we demonstrate task assignment and progress monitoring for the **CA Firm Attendance and Article Management System**. Because this project involves complex dual-verification logic (Face Recognition + GPS) and distinct user portals (Admin vs. Article Trainee), we simulate the development lifecycle using two industry-standard tools:

1. **JIRA (Scrum Approach):** Used for managing the core backend structure, database, and biometric API integrations within 2-week Sprints.
2. **Trello (Kanban Approach):** Used for managing the React.js UI components and maintaining a continuous flow of front-end deliverables.

3. JIRA Implementation (Scrum Framework)

Jira Software by Atlassian is utilized to manage the intensive backend logic (Node.js/MySQL) and security constraints of the CA Firm System.

3.1 Project Setup & Hierarchy

- **Project Name:** CA Firm Backend Integration (CABI)
- **Epics:** Large bodies of work broken down into stories.
 - *Epic 1:* Biometric Face verification module (face-api.js)
 - *Epic 2:* Administrative Gamification & Reporting
 - *Epic 3:* Location Services (GPS/Haversine logic)

3.2 Sprint 2 Planning (Simulated)

Sprint Goal: Complete the dual-factor attendance backend verification (GPS + Face matching).

Issue Key	Type	Summary	Assignee
CABI-201	Story	Write Haversine GPS radius calculation logic	Nivesh J.
CABI-202	Story	Integrate face-api.js euclidean distance compare	Harsh N.
CABI-203	Task	Set up MySQL Attendance tracking table	Rachit I.
CABI-204	Bug	Fix HTTP 500 on missing location permission	Hasan R.
CABI-205	Task	Write unit tests for attendance success/failure	Nivesh J.

3.3 Jira Workflow States

Issues in the active sprint board follow a strict status flow: **To Do** → **In Progress** → **Code Review** → **Done**.

[Insert Jira Sprint Board Screenshot Here]

Instructions: Create a free Jira project, add the 5 tasks listed above to an active sprint, move a few to "In Progress", and paste the screenshot here.

4. Trello Implementation (Kanban Framework)

Trello is utilized for the frontend React.js development of the Admin and Article Dashboards.

4.1 Board Layout (Lists)

Our Trello board represents a continuous delivery pipeline:

1. **Backlog:** All planned UI views (e.g., Leave Application Form).
2. **Up Next:** Priority components to be coded this week.
3. **In Development:** Components currently being coded.
4. **Testing/Staging:** Components under peer review.
5. **Done:** Successfully merged code.

4.2 Detailed Card Simulation

Each Trello card represents a specific component containing checklists and tags.

Card Title: *Admin Dashboard - Task Assignment Form*

- **Assignee:** Hasan Rupawalla
- **Labels:** [Frontend], [Admin View], [Needs API Connection]
- **Checklist (Definition of Done):**
 - ☐ Create form fields (Title, Desc, Deadline, Trainee Dropdown)

- ☐ Add validation for required fields
- ☐ Connect to `POST /api/tasks` endpoint
- ☐ Display success/error toast notification

[Insert Trello Kanban Board Screenshot Here]

Instructions: Create a free Trello board with the lists mentioned above. Add a few cards (like "Admin Dashboard", "Login Page", "Camera component"), assign them, and add labels. Paste the screenshot here.

5. Conclusion

Through this exercise, we successfully simulated task assignment and progress tracking for the **CA Firm Attendance System**. By leveraging **JIRA** for backend sprints involving delicate API integrations and **Trello** for visual, Kanban-style frontend component tracking, the development process becomes structured and transparent. This dual-tool approach ensures that deep logic and fast-paced user interface design are managed according to their respective agile needs.